Alexander Dao

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Education

University of Florida Aug. 2023 – May 2027
Bachelor of Science in Computer Science, Minor in Japanese, GPA: 3.8 / 4 *Gainesville, FL*

Kansai Gaidai University Aug. 2024 – June 2025
Asian Studies and Business Program *Osaka, Japan*

Palm Beach State College Aug. 2020 – May 2022
Associate of Arts *Lake Worth Beach, FL*

Experience

IBM TRIRIGA Senior Technical Analyst III Aug. 2022 – July 2023
Palm Beach County School District *West Palm Beach, FL*

- Developed new IBM TRIRIGA Business Objects for calculating Work Order requirements
- Converted legacy desktop interface into offline capable mobile web application with React for school technicians working in remote facilities
- Corresponded with school technicians to digitize scheduling forms into the TRIRIGA database

IT Project Management Office Intern June 2022 – July 2022
Palm Beach County School District *West Palm Beach, FL*

- Automated converting **800+** embedded school and phone record data into an index-able spreadsheet utilizing AppleScript and Vimium condensing archival teams work schedule in half
- Developed the intern program schedule, including weekly events, project timelines, and follow-up survey, to adapt to changing vendor and district administration's availability, facilitating an uninterrupted weekly schedule

Projects

HomeLab | *Docker, NixOS, Kubernetes, SolidWorks* Sep. 2023 - Present

- Centralized configuration with **NixOS** with feature modules utilizing Dendritic data pipeline, allowing for scalable deployment and version rollbacks with **GitOps** and **Jujutsu**
- Maintained up-time with containerized **Docker** application deployed with **Kubernetes** across multiple locations in case of power outages, with attached IPKVM to allow for remote desktop access and recovery
- Secured all data with a 3-2-1 backup convention and data verification via ZFS file format
- Operated private cloud and media network to be independent from large cloud providers, providing services such as Jellyfin, Syncthing, NextCloud, SSH, and SFTP along with a custom router using **OpenWrt**
- Collaborated with **Claude Code** to generate CAD drawings with manual optimizations via **Solidworks** for custom 3D-printed server components

DoomCloud | *Lisp, Emacs, Org-Roam, Docker* Jan. 2024 - Present

- Engineered an interactive web portal for a containerized custom-built **Emacs** TUI binary to publish and enable shareable Lisp macros for non-technical users.
- Architectured a **200+** page Zettelkasten style note organization system along with GTD tasks system to standardize information ingest and project startup process

Command Line Image Editor | *Rust* Feb. 2024

- Developed CLI tool to apply mathematical function upon pixel data within a larger data pipeline
- Executed Test Driven Development with red, green unit tests ensuring application followed specification

Stats of America | *Go, React* May 2024

- Designed parallelize-able data pipeline with **Go**-runtime to ingest US governmental data with **8x** speed uplift
- Collaborated asynchronously via GitHub within an agile team to create a website displaying correlated trends within data.gov **300,000+** CSV dataset

Technical Skills

Relevant Coursework: Data Structure & Algorithms, Operating Systems, Competitive Programming
Languages and Frameworks: Rust, C++, Go, Emacs-Lisp, Bash, Python3, JavaScript
Developer Tools: Nix, Colmena, Kubernetes, Docker, Emacs, Org-Roam, Org-Tangle, Git, Jujutsu, SQLite
Certificates: Comptia A+, Biotechnician Assistant Credentialing Exam, CSWA CAD Certification